

rotation. Taking this into account, the program calculates the RPM value and displays it on the LED module.

The speed.ino program uses Timer-1 for counting seconds. The INT-0 interrupt on pin 2 is triggered for every pulse received from the sensor. Each pulse invokes an interrupt, and the interrupt service routine (ISR) for INT-0 counts every pulse.

Every eight seconds, the pulse count value is read and stored. After

```
#define ir 2
boolean ir1=(digitalRead(ir));

void setup()
{
  pinMode(ir, INPUT);

  pinMode(2, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  ir1= (digitalRead(ir));
  Serial.println(ir1);
}
```

Fig. 5: Code snippet of the main program

```
39
40
41 ISR(TIMER1_OVF_vect){
42   TCNT1=0;
43   digitalWrite(13,digitalRead(13)^1);
44   if (n>2)
45   {n=0;
46    n1=n1*7.5/4; // (8 seconds x 7.5 = 1min. divide by 3 for 3 blade fan or 4 for fourblade fan)
47    n11=n1/1000;
48    n12=n1-1000*n11;
49    n13=n12/100;
50    n14=n12-100*n13;
51    n15=n14/10;
52    n16=n14-10*n15;
53
54    write(digits[n11], digits[n13] , digits[n15], digits[n16]);
55    Serial.println(n1);
56    n1=0;
57   }
58   n=n+1;
59   }
60 void loop() {
61   }
62 void write(uint8_t first, uint8_t second, uint8_t third, uint8_t fourth)
63 {
64   start();
65   writeValue(0x40);
66   stop();
67   start();
68   writeValue(0xc0);
69   || writeValue(first);
70   //writeValue(first);
71   writeValue(second);
72   writeValue(third);
73   writeValue(fourth);
74   stop();
75 }
```

Fig. 6: Code snippet converting pulse to speed

this, the count is reset to zero to start the next measurement cycle.

The stored value is then converted to RPM by dividing it by 4 and multiplying by the required factor to convert the reading to per minute. The calculated value is then sent to the LED display.

The LED display requires initialisation, which is done using the functions start(), stop(), and write(), as shown in Fig. 6.

Construction and testing

After downloading the speed.ino program from the link, save the file on the computer and open it in the Arduino IDE.

Then select the the Arduino Uno board from the Tools menu and choose the correct port to which the Arduino board is connected. After confirming these settings, click Upload to transfer the code to the Arduino Uno.

Next, assemble the circuit on a small general-purpose PCB or

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breadboard, as shown in the prototype in Fig. 1.

The device can be configured to measure speed in various scenarios. For example, to measure the speed of a ceiling fan, a rod can be used to hold the IR module at its tip using tape. The rod is then kept near the fan.

The three wires connected to the module are extended using thin, flexible wires. This can be done by cutting the Arduino female-to-male jumper wires and inserting additional wires in between. At least 3-metre wire length is required.

If a two-core cable, such as an audio cable, is used, the arrangement becomes more convenient.

The rod should extend about 1cm beyond the IR module to ensure that the LEDs on the module do not hit the fan blades.

Fig. 7 shows the ceiling fan speed measurement using a rod with the IR module mounted at the top and held near the blades.

The reading must also be adjusted



Fig. 7: Ceiling fan speed measurement using a rod with an IR module held near the blades